Gen AI Developer - Practical Technical Assessment

Assignment: AI-Powered SVG Banner & Infographic Generation Engine

1. Objective

At Shri Genesis Software Solutions, we work on AI/ML, Generative AI, automation, and scalable digital solutions.

As part of this assessment, you are required to develop a small AI-powered application that automatically generates professional SVG banners and infographics for different topics and categories.

The solution should combine Generative AI, content generation, layout design, vector graphics, and automation to produce visually appealing and reusable SVG designs.

The objective is not simply to generate SVG files, but to build a system capable of producing original, professional, clean, and production-quality visual designs.

2. Core Assignment

Build a script, application, or workflow that generates approximately 100 SVG banners or infographic designs across different topics and categories.

The application should be capable of automatically generating:

- Topic or category
- Banner or infographic concept
- Written content and copy
- Headings and subheadings
- Supporting text
- Icons or illustrations
- Color combinations
- Typography
- Layout and positioning

- Final SVG output

The system should generate designs programmatically rather than requiring each banner to be manually created.

Example Categories

You may select any suitable categories, such as:

- Technology and AI
- Business and startups
- Education
- Healthcare
- Finance
- Marketing
- Cybersecurity
- E-commerce
- Real estate
- Productivity
- Sustainability
- Social media awareness
- Corporate announcements

You may use other categories if they demonstrate the flexibility of your solution.

3. Functional Requirements

Your application should support the following capabilities.

A. Topic and Content Generation

The system should:

- Accept a topic, category, or content brief.
- Generate suitable banner or infographic content.
- Create relevant titles, subtitles, descriptions, and key points.
- Maintain logical content hierarchy.
- Avoid irrelevant, repetitive, or grammatically incorrect content.

B. Template and Layout Generation

The system should generate multiple design layouts, such as:

- Promotional banners
- Educational infographics
- Statistics-based designs
- Step-by-step infographics
- Quote banners
- Announcement banners
- Comparison layouts
- Feature highlights
- List-based infographics
- Corporate or social media graphics

The output should not consist of the same layout with only the text or colors changed.

C. SVG Generation

Each output must be a valid, scalable SVG file and should include:

- Proper dimensions and viewBox
- Text and typography
- Shapes and vector elements
- Icons or illustrations where relevant
- Proper spacing and alignment
- Suitable color contrast
- Responsive scaling where applicable
- Clean and organized SVG structure

The SVG files should be viewable in standard browsers and common vector-design tools.

D. Variety and Reusability

The application should demonstrate:

- Multiple templates
- Multiple visual styles
- Different content structures
- Different compositions and layouts
- Reusable design components
- Configurable colors, fonts, spacing, and dimensions

The system should be designed so that additional categories and templates can be added without major code changes.

4. Mandatory Design Authenticity, Originality & SVG Quality Requirements

This section is an essential part of the assignment.

The generated designs must demonstrate human-directed design decisions, originality, professional visual quality, and clean technical implementation.

A. Original and Professional Design

The designs should:

- Have intentional and well-balanced layouts.
- Use appropriate typography and visual hierarchy.
- Maintain consistent spacing and alignment.
- Avoid generic, repetitive, or visibly unfinished AI-generated compositions.
- Demonstrate meaningful design variation across outputs.
- Use visual elements that are relevant to the topic.
- Avoid excessive decorative elements that do not add value.
- Be suitable for professional, commercial, or marketing use.

The objective is to create designs that look like they were developed through a thoughtful design process rather than produced through uncontrolled or repetitive generation.

B. Clean SVG Code and Metadata

The generated SVG files must have clean and maintainable code.

Ensure that the SVG output does not contain:

- Unnecessary AI prompts or generation instructions.
- Model-specific metadata that is irrelevant to the final design.
- Unnecessary AI-generated comments.
- Hidden or unrelated elements.
- Invisible text or misleading content.
- Unnecessary external dependencies.
- Tracking scripts or suspicious embedded code.
- Excessive duplicate elements.
- Broken or unused definitions.
- Fabricated authorship or provenance claims.

The SVG code should be:

- Readable and reasonably organized.
- Structured using reusable elements where appropriate.
- Free from unnecessary technical artifacts.
- Easy for another developer to inspect and maintain.
- Suitable for further editing or customization.

Important: Cleaning metadata and SVG code is a quality and maintainability requirement. It must not be used to make false claims about the origin of the design or to mislead reviewers about the tools used.

C. Originality and Intellectual Property Compliance

All designs must be original or created using assets that you are legally authorized to use.

You must not:

- Copy designs from existing websites, brands, templates, or designers.
- Reproduce copyrighted illustrations or infographics without permission.
- Use unauthorized brand logos, trademarks, or proprietary graphics.
- Use stock icons, fonts, images, or illustrations without checking their licensing terms.
- Recreate an existing design with only minor modifications.
- Include content or assets that violate third-party intellectual property rights.

You may use:

- Original SVG illustrations created by you.
- Open-source assets with appropriate licenses.
- Properly licensed icon libraries.
- AI-generated assets where their use is permitted.
- Self-created design components and templates.

Please mention the libraries, fonts, datasets, APIs, and AI tools used in your submission.

D. AI Detection and Design Review

The submitted designs may be reviewed using tools such as ChatGPT, Claude, or other design-analysis systems.

The expectation is that the output should demonstrate:

- Original design decisions.
- Clean and professional SVG structure.
- No unnecessary AI-related artifacts.
- Meaningful variation between designs.
- Proper typography and layout.
- No unauthorized copying or reused copyrighted designs.
- Transparent documentation of the development process.

However, no AI system can reliably guarantee that a design will be classified as "human-designed" or "not AI-generated." Therefore, candidates will not be evaluated solely on the result of an AI detector.

Candidates must not make false claims regarding authorship, design origin, or the tools used. The focus will be on actual design quality, originality, technical cleanliness, and responsible use of AI.

5. Generative AI and Technical Expectations

You may use suitable technologies and AI services, including:

- OpenAI APIs
- Other LLM APIs
- LangChain or similar frameworks
- Python
- Node.js
- JavaScript or TypeScript
- SVG libraries
- HTML/CSS-based rendering
- Vector-generation libraries
- Image or icon generation tools
- Prompt templates
- Structured JSON-based generation
- Rule-based layout engines

The solution should demonstrate how AI is used meaningfully in the workflow.

Examples include:

- Generating content based on a topic.
- Selecting a suitable design template.
- Creating layout instructions.
- Generating color and typography suggestions.
- Producing structured SVG elements.
- Validating and refining the generated output.
- Detecting layout or content issues.
- Generating multiple variations automatically.

Simply calling an AI API and saving unstructured responses as SVG files will not be sufficient.

6. Output Requirements

Your application should generate approximately 100 SVG outputs across multiple topics and categories.

The outputs should include:

- SVG files
- Relevant topic or category
- Design variation
- Generated content
- Consistent file naming
- Organized folder structure

You may generate the files in batches using reusable templates and automated workflows.

Submission Volume

Please submit:

1. Approximately 100 generated SVG files.
2. A representative preview set of 10-15 designs showing the range of layouts and categories.
3. A contact sheet or preview document, if possible.
4. Source code and configuration files.
5. A short technical report.

The 100 outputs may be generated programmatically, but they should not be simple duplicates with minor text or color changes.

7. Suggested Input and Output Structure

You may use a structured input format similar to the following:

```
{
  "category": "Technology",
  "topic": "Benefits of Artificial Intelligence",
  "design_type": "Infographic",
  "style": "Modern Corporate",
  "tone": "Professional",
  "color_theme": "Blue and White",
  "output_format": "SVG"
}
```

An example output structure could be:

```
{
  "id": "design_001",
  "category": "Technology",
  "topic": "Benefits of Artificial Intelligence",
  "design_type": "Infographic",
  "template": "feature_highlight",
  "svg_file": "outputs/design_001.svg",
  "content_generated": true,
  "validation_status": "passed"
}
```

This is only an example. You may use a different structure based on your implementation.

8. Validation and Quality Checks

Your application should include reasonable validation wherever possible.

Consider validating:

- Whether the SVG is syntactically valid.
- Whether the SVG contains a valid `viewBox`.
- Whether required elements are present.
- Whether text exceeds its designated area.
- Whether elements overlap unexpectedly.
- Whether colors have sufficient contrast.

- Whether fonts and external assets are available.
- Whether the output contains broken references.
- Whether duplicate designs are being generated.
- Whether the generated content is complete and relevant.

You may include an automated quality-checking process or a manual review report.

9. Submission Requirements

Please submit the following:

1. Source Code

- Complete source code.
- Configuration files.
- Dependency file such as `requirements.txt` or `package.json`.
- Environment variable documentation.
- Instructions for running the application.

Do not include private API keys or credentials.

2. Generated SVG Files

- Approximately 100 SVG files.
- Organized folders by category, topic, or batch.
- A representative preview of the generated designs.

3. Technical Report

Submit a short report of approximately 2-4 pages covering:

- Solution overview.
- Technology stack.
- AI models or APIs used.
- Prompting approach.
- Template and layout-generation approach.
- SVG generation process.
- Originality and licensing approach.
- Metadata and SVG-cleaning approach.
- Validation process.
- Challenges faced.
- Possible future improvements.

4. Execution Instructions

Explain:

- How to install dependencies.
- How to configure API keys.
- How to provide input topics.
- How to generate the SVG files.
- How to generate approximately 100 outputs.
- How to validate or preview the output.

5. Execution Report

Include:

- Total number of designs generated.
- Number of successful outputs.
- Number of failed or partially generated outputs.
- Total execution time.
- Average generation time per design.
- API or model usage, if applicable.
- Validation results.
- Any known limitations.

10. Suggested Time Commitment

The assessment is intended as a preliminary technical evaluation, not a complete production system.

Suggested effort:

Approximately 4-8 hours, depending on the candidate's approach and the level of automation implemented.

Candidates are not expected to build a fully commercial-ready platform. The submission should demonstrate:

- Technical understanding.
- Practical Generative AI implementation.
- SVG and vector-design knowledge.
- Code quality.
- Problem-solving ability.

- Automation skills.
- Attention to originality and visual quality.

11. Evaluation Criteria

Evaluation Area	What We Will Review
Generative AI Implementation	Meaningful and practical use of AI
SVG Generation	Valid, scalable, and technically correct SVG files
Design Quality	Typography, spacing, alignment, hierarchy, and visual appeal
Design Authenticity	Original layouts and meaningful design decisions
Design Variety	Different categories, templates, and compositions
Code Quality	Readable, modular, maintainable implementation
SVG Cleanliness	Organized code, clean metadata, and no unnecessary artifacts
Originality	Original designs and responsible asset usage

IP Compliance	Proper licensing and avoidance of unauthorized content
Automation	Ability to generate multiple outputs efficiently
Validation	Handling of errors, broken layouts, and invalid SVGs
Documentation	Clear setup instructions and technical explanation

12. Important Guidelines

- Do not share private API keys, credentials, or confidential information.
- Do not use unauthorized data, copyrighted templates, or proprietary assets.
- Do not bypass security controls, access restrictions, or platform protections.
- Clearly disclose the AI tools, libraries, and external assets used.
- Ensure that the generated designs are original and suitable for review.
- The application should be developed with maintainability and scalability in mind.
- The submitted code and assets must be created or used in accordance with applicable licenses and laws.

Final Candidate Checklist

Before submitting, please ensure that you have completed the following:

- Developed an application or script for automated SVG generation.
- Implemented AI-based content or design generation.
- Included multiple topics and categories.
- Implemented multiple templates or layouts.
- Generated approximately 100 SVG outputs.
- Included a representative preview of the designs.
- Ensured the SVG files are valid and scalable.
- Checked typography, alignment, spacing, and layout quality.
- Removed unnecessary metadata and SVG code artifacts.

- Ensured the designs are original and not copied.
- Verified the licensing of fonts, icons, and other assets.
- Included source code and setup instructions.
- Submitted the technical and execution reports.
- Disclosed the AI tools and external resources used.

The primary objective is to demonstrate your ability to build a practical Generative AI solution that produces original, professional, scalable, and technically clean SVG banners and infographics through automation.